

REQUIREMENT ANALYSIS

Customer Journey Map

Project Title	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Repository	https://github.com/Alisha40477/Electricity-Consumption-Analysis
Website	Flask Web Application with Embedded Tableau Dashboard
Team Members	Mohammed Mahaboob Alisha Shaik, Hemanth Kallepalli, Satish Vujjuri, Sudheeksha Sangana, Veligatla Pavan Kumar

Project Description

The project analyzes electricity consumption data collected from different states and regions using **MySQL**, **Tableau Desktop**, **Tableau Public**, and **Flask**. It provides interactive dashboards and stories that help users understand electricity usage patterns and support better energy planning.

The dashboard contains:

- KPI Dashboard
- State-wise Electricity Usage
- Region-wise Electricity Usage
- Monthly Electricity Usage Trend
- State-wise Electricity Usage Map
- Top 10 Electricity Consumers
- Interactive Filters
- Business Insights & Conclusion

The objective of the project is to help electricity departments, government organizations, energy analysts, and researchers understand electricity consumption patterns and make informed decisions through interactive visualizations.

Customer Persona

Primary User

Energy Analyst

Age

25–50 Years

Occupation

Energy Analyst / Government Electricity Officer

Experience

Responsible for monitoring electricity consumption, preparing analytical reports, studying regional electricity demand, and supporting energy planning through data analysis.

Goals

- Monitor electricity consumption.
- Compare electricity usage among states.
- Analyze regional electricity demand.
- Study monthly electricity trends.
- Identify high electricity-consuming states.
- Support data-driven decision making.

Challenges

- Large electricity datasets.
- Manual report preparation.
- Difficulty identifying trends.
- Lack of interactive dashboards.
- Time-consuming analysis.

CUSTOMER JOURNEY MAP

❖ Stage 1 — Access Dashboard

User Goal

Access electricity consumption dashboard.

Customer Experience

The user opens the Flask application and accesses the Tableau dashboard.

Touchpoints

- Flask Web Application
- Tableau Dashboard

Positive Experience

Dashboard loads successfully.

Pain Point

New users may not understand dashboard navigation.

Opportunity

Provide dashboard introduction and navigation guide.

❖ Stage 2 — Explore Electricity Data

User Goal

Understand electricity consumption.

Customer Experience

The user views KPIs and electricity usage charts.

Touchpoints

- KPI Dashboard
- State-wise Dashboard
- Region-wise Dashboard

Positive Experience

Interactive charts provide quick understanding.

Pain Point

Large amount of information.

Opportunity

Provide tooltips and chart descriptions.

Stage 3 — Analyze Trends

User Goal

Identify electricity usage trends.

Customer Experience

The user applies filters to analyze states and regions.

Touchpoints

- Filters
- Maps
- Trend Charts

Positive Experience

Dashboard updates instantly.

Pain Point

Selecting multiple filters may confuse beginners.

Opportunity

Add Reset Filter button and user instructions.

❖ **Stage 4 — Generate Insights**

User Goal

Understand business insights.

Customer Experience

The user reviews Tableau Story and Business Insights.

Touchpoints

- Tableau Story
- Business Insights Page

Positive Experience

Story explains findings clearly.

Pain Point

Requires updated datasets.

Opportunity

Enable regular dashboard refresh.

❖ **Stage 5 — Decision Making**

User Goal

Use dashboard insights for planning.

Customer Experience

The user prepares reports and supports electricity planning.

Touchpoints

- Dashboard
- Reports
- Tableau Public

Positive Experience

Supports faster decision making.

Pain Point

Future prediction is unavailable.

Opportunity

Integrate AI-based electricity demand forecasting.

Customer Journey Summary

Stage	User Goal	Positive Experience	Pain Point	Opportunity
Access Dashboard	Open dashboard	Easy access	Navigation confusion	User guide
Explore Data	Understand usage	Interactive charts	Large information	Tooltips
Analyze Trends	Compare states	Dynamic filters	Filter complexity	Reset filters
Generate Insights	View reports	Story explains results	Needs updated data	Automatic refresh
Decision Making	Plan electricity usage	Better planning	No forecasting	AI prediction

Conclusion

The Customer Journey Map helped the project team understand how users interact with the Electricity Consumption Analysis system from accessing the dashboard to making informed decisions. It identified user goals, challenges, experiences, and improvement opportunities at every stage.

The insights gained from the journey map guided the development of an interactive solution using **MySQL, Tableau Desktop, Tableau Public, and Flask**. The project enables users to analyze electricity consumption efficiently, compare state-wise and region-wise usage, monitor monthly trends, and generate meaningful business insights.

The Customer Journey Map ensures that the developed system is user-friendly, supports effective energy planning, improves the overall user experience, and promotes data-driven decision-making.